Joshua Sternadel

Kansas City, MO | Contact via LinkedIn
jsternadel.github.io/portfolio/

Principal Systems & Software Engineer

Core Competencies: Full-Stack Architecture (Angular/Node/.NET), Infrastructure Automation, & High-Availability Systems
U.S. Patent Author (#10,366,069) – Database Management & Natural Sorting Algorithms
Enterprise Scale: 16 Years Systems Engineering | Tech Anchor for Fortune 500 Telecom Mergers

Key Achievements

- **Strategic Retention & Continuity:** Awarded a significant retention bonus during the Sprint/T-Mobile merger, designated as mission-critical personnel to ensure technical stability and architectural integration during the corporate transition.
- **U.S. Patent Author (#10,366,069):** Co-authored a patent for database management systems, specifically engineering a natural sorting algorithm in SQL to optimize data retrieval and user experience.
- **Infrastructure Transformation:** Orchestrated a full-scale migration from legacy physical servers to a redundant VMware-based virtual infrastructure, reducing enterprise downtime by over 90% and cutting operational payroll costs by 18%.
- **Algorithmic Solutions:** Engineered a first-principles solution to resolve GPS signal noise and "teleportation" errors for intermodal logistics. By deciphering raw NMEA 0183 sentences and applying spherical geometry with rate-of-acceleration modeling, successfully stabilized coordinate paths and eliminated tracking jitter within high-interference rail yards.

Professional Experience

T-Mobile | Principal Solutions Architect & DevOps Engineer | 2021 to 2026

- **Cross-tenant Enterprise Cloud Migration:** Architected and executed the zero-loss migration of 10+ years of metadata, complex workflows, and device certification data from Sprint's Jira Data Center to T-Mobile's Jira Cloud in just 3 months. Deployed a custom PHP translation framework to route and sync complex schema across systems, maintaining zero operational downtime for 3,000+ active enterprise users.
- **Tooling Reuse & Ecosystem Dashboard:** Ported custom framework tooling and hydration engines to build internal tool ecosystem dashboards. Reduced boilerplate code by 10x and eliminated development overhead across internal application environments.

- **Automated Incident Integration Pipeline:** Authored custom fluent SDKs in Node.js and PHP for Atlassian APIs, creating custom Node-RED nodes and REST endpoints. Integrated this pipeline with Ansible to capture deployment errors, parse JSON payloads, and automatically generate structured Jira tickets for real-time incident tracking.
- **Infrastructure as Code (IaC):** Bridged the gap between Software Engineering and Systems Administration by implementing Infrastructure as Code (IaC) via Ansible, successfully reducing manual operational toil and human error.
- **DevSecOps & Automation:** Engineered automated vulnerability remediation workflows that drove lab data center security debt from 100+ monthly criticals to zero, achieving a state of continuous compliance.
- **Value Stream Optimization:** Optimized the VM lifecycle by automating the end-to-end onboarding process, significantly accelerating the “Lead Time to Deployment” for internal engineering teams.
- **Full-Lifecycle Application Engineering (Angular & Serverless):** Authored and launched a high-visibility 5G Speed Calculator application utilizing Angular and a modern serverless deployment pipeline to ensure instant scalability and zero infrastructure overhead.
- **UI/UX Optimization:** Bridged the gap between UI/UX design concepts and technical feasibility, establishing strict component styling and layout standards to ensure frontend performance and visual consistency.

Sprint | Principal Engineer & Application Developer III | 2017 to 2021

- **Executive Technical Governance:** Served as Core Technical Advisor and Architectural Gatekeeper to leadership; held final engineering sign-off authority to halt high-risk regressions across all cross-functional integrations.
- **Full-Stack Architectural Execution:** Handed a critical enterprise SKU bottleneck that had suffered through a decade of failed attempts. Took complete ownership of the solution strategy and architecture, single-handedly engineering 100% of the frontend application layer while orchestrating the underlying Porcupine message bus to deliver a high-performance production system in just 6 months.
- **Distributed Systems:** Designed and deployed “Porcupine,” a custom event-driven message bus architected on top of RabbitMQ; seamlessly synchronized live Jira workflow states with downstream application environments, establishing a single source of truth and eliminating cross-system data lag.
- **Custom Data Framework & Universal Hydration Engine:** Engineered a data-agnostic PHP framework with a custom hydration engine using reflection and annotations. Treated disparate data formats (SQL, NoSQL, JSON, XML, HTML) as unified entity-managed ORM models—abstracting storage-layer complexity and reducing boilerplate code by 10x for new service onboarding.

- **Scale & Adoption:** Prototyped and deployed the framework to internal high-value T-Mobile applications, proving a 10x reduction in “boilerplate” code for new service onboarding.
- **Data Architecture & Prototyping:** Architected and validated a standalone Dremio data lakehouse POC designed to federate data layers across distributed MS SQL, MySQL, and MongoDB environments. Proved the viability of zero-ETL query federation across relational and NoSQL silos, establishing a blueprint for unified data access.
- **Observability & Infrastructure Telemetry:** Built an integrated observability platform using the ELK Stack, Prometheus, and Grafana across a 45-VM development environment.
- **Incident Response Systems:** Designed end-to-end infrastructure telemetry and automated alert routing hooks via Rocket.Chat to demonstrate proactive incident response frameworks.
- **Real-Time Telemetry:** Engineered a custom heartbeat monitoring framework embedded within distributed .NET Core applications; piped real-time telemetry directly into Grafana to ensure proactive incident response and eliminate production blind spots.
- **Financial Stewardship & Capital Forecasting:** Managed and optimized the departmental infrastructure and licensing budget as it scaled from an annual spend of $300K to $1.3M. Formulated highly accurate 12-month financial and capacity projections to align capital expenditures with rapid system expansion.

Sprint | Product Development Engineer II | 2013 to 2017

- **Systems-Level Engineering Transition:** Capitalized on hardware validation background to bridge the gap between device testing and robust application development, driving strict performance constraints into early-stage software architecture.
- **Full-Stack Application Architecture:** Conceived, architected, and built a visual drag-and-drop prototyping platform to simulate Out-of-Box Experiences. Integrated custom telemetry engines directly into the app to capture real-time CPU, RAM, and battery metrics for device validation.
- **Database Engineering & Tooling:** Authored the foundational codebase for the Device Development and Certification requirements and test-plan database, laying the groundwork for scalable documentation and testing.
- **Cross-Functional Engineering Governance:** Served as the authoritative technical liaison to the Head of Retail Testing and executive leadership; directed weekly technical reviews to deconflict complex system behaviors, prioritize software bug remediation, and establish release readiness timelines for the nationwide SLATE ecosystem.

EXPERIS (Contractor at Sprint) | Technical Support Rep II | 2012 to 2013

- **Performance Automation Overhaul:** Re-engineered and optimized a legacy device performance tracking application used across the ecosystem.

- **Nationwide Enterprise Automation Deployments:** Engineered and maintained the core XML execution scripts driving the SLATE diagnostic framework deployed across Sprint's entire national retail retail footprint; ensured 100% flawless execution of automated on-site device validation routines for in-store service centers.

MO-KAN Container Services | System Administrator | 2011 to 2012

- **Multi-Site Infrastructure Governance:** Directed end-to-end technology operations across 3 distributed corporate locations; modernized legacy environments by deploying virtualized, redundant infrastructures that cut enterprise downtime by 90%.

Additional Relevant Experience

Trucking Central | Software Consultant (Contract): Planned, developed, and tested Android-based GPS solutions for the intermodal supply chain industry.
Innovative Solutions | Mobile Solutions Technician: Troubleshot and fixed Nextel devices for first responders and construction businesses, listened to client needs, and leveraged emerging technologies to find solutions for those needs.

Community Impact & Leadership

Seek and Serve | Technical Advisory Board Member | 2026 to Present

- **Technical Guidance:** Provide high-level architectural oversight and logistical strategy to the parent 501(c)(3) organization.

33meals.org (Subsidiary) | Infrastructure Consultant & Volunteer | 2026 to Present

- **Organizational Launch:** Spearheaded the launch and incorporation of 33meals.org as a dedicated subsidiary to address food insecurity through scalable, data-driven, distribution models.

- **Operational Infrastructure:** Managed the full technical and administrative setup, including domain registration, digital architecture, and regulatory filings to ensure high-integrity non-profit operations.

- **Strategic Development:** Designed a streamlined logistics model to optimize charitable food delivery, applying enterprise efficiency principles to humanitarian service.

Education

- **Associate of Arts (AA), Graphic Design** – Johnson County Community College

Technical Environment Matrix

- **Architectural Disciplines:** Systems Architecture, Distributed Systems, Microservices, API Gateway Design, Platform Abstractions, SRE, DevSecOps
- **Frontend Web Frameworks:** Angular (Pre-1.0 to v15), Vue.js 2, TypeScript, JavaScript (ES6+), Client-Side Architecture
- **Backend Environments & Languages:** Node.js Runtime, .NET Core (2.x), C#, Native Modules, Zero-Dependency Design
- **Infrastructure & Automation:** Ansible, DevOps, CI/CD, Infrastructure as Code (IaC), Automation, Site Reliability, Serverless, Red Hat Linux, VMware
- **Data & Ecosystems:** Dremio (Lakehouse Federation), ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Rocket.Chat, PostgreSQL, SQL, MongoDB, Atlassian SDK, Jira API (US Patent #10,366,069)